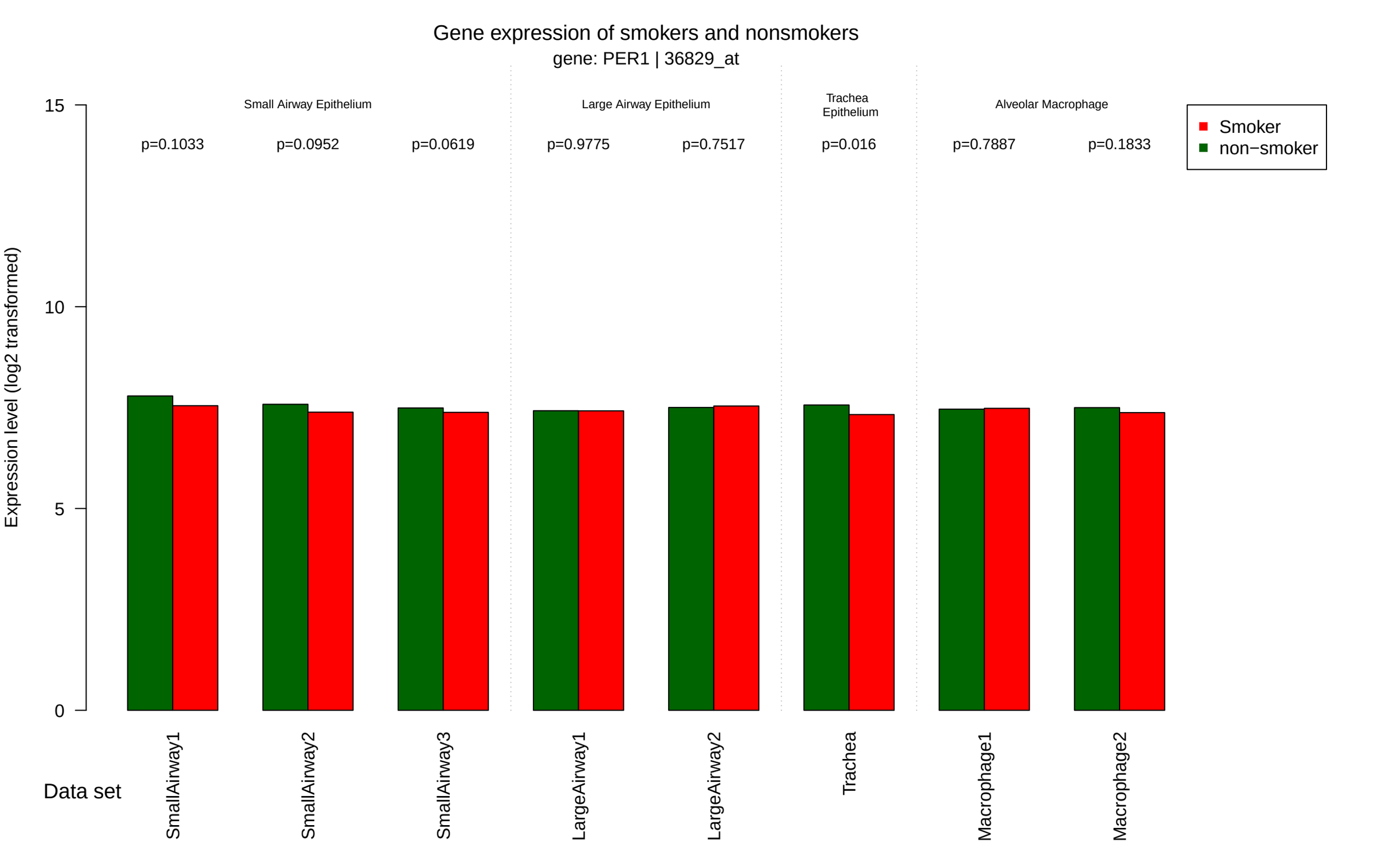
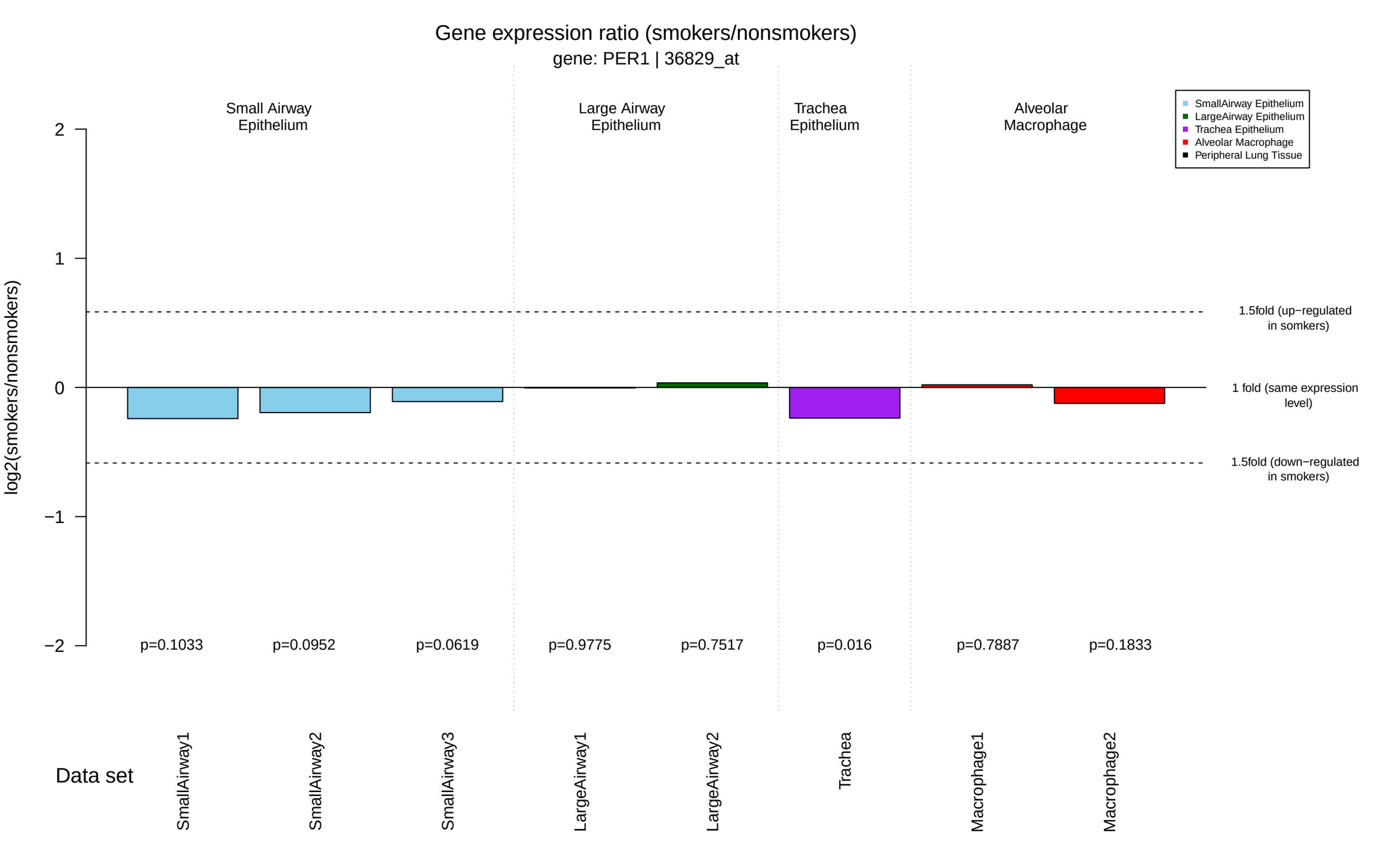
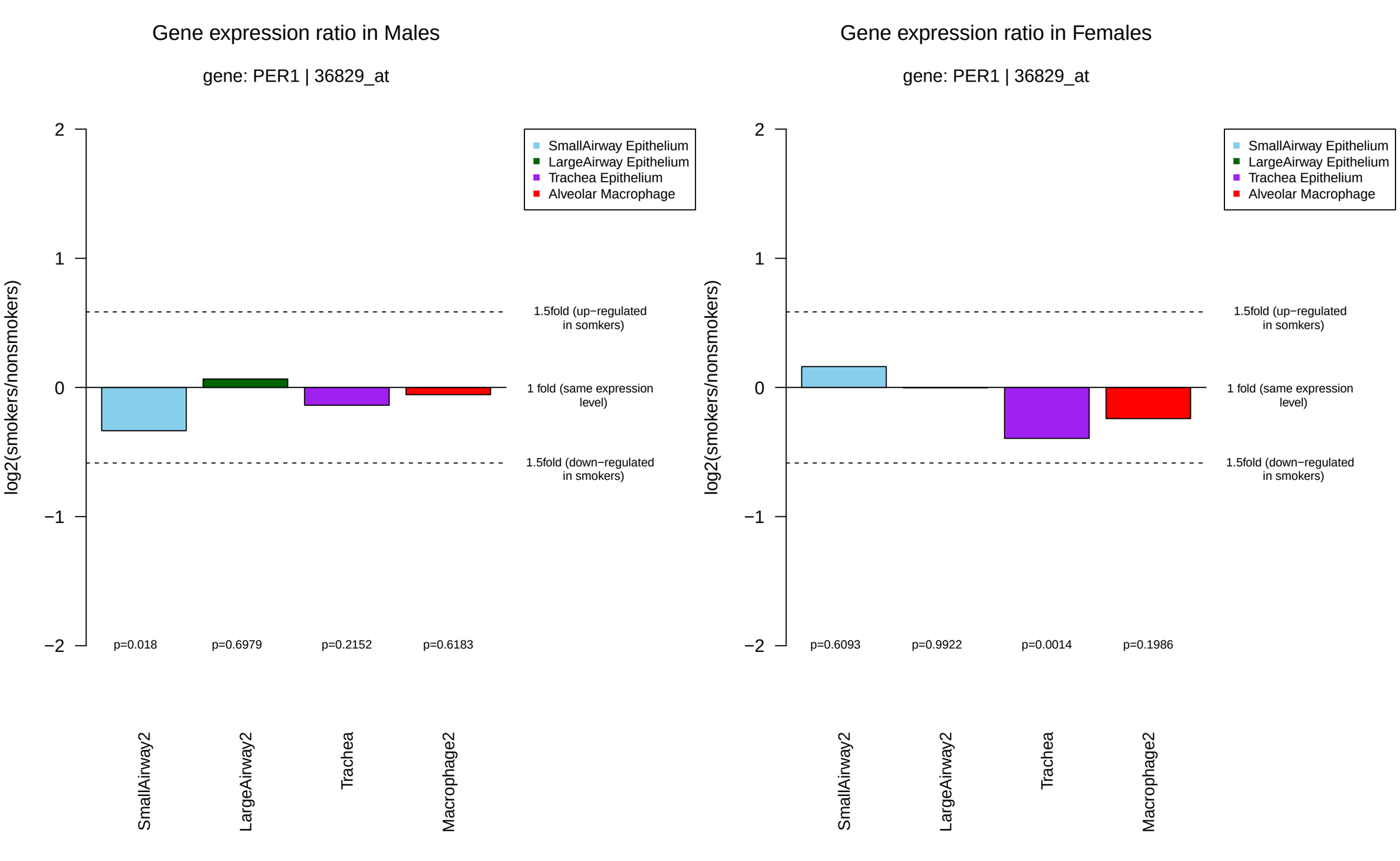
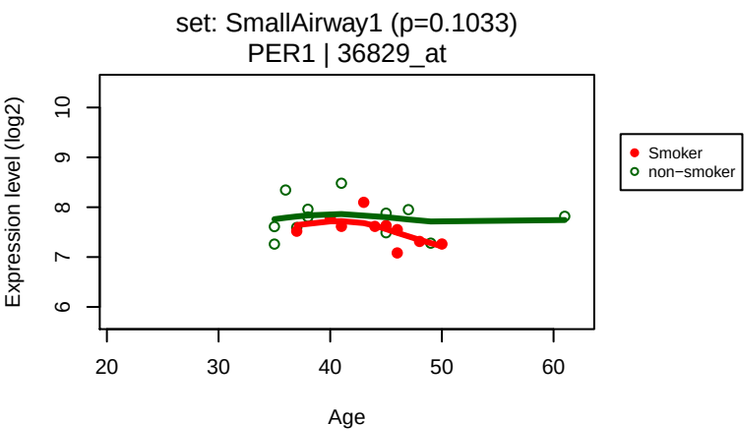
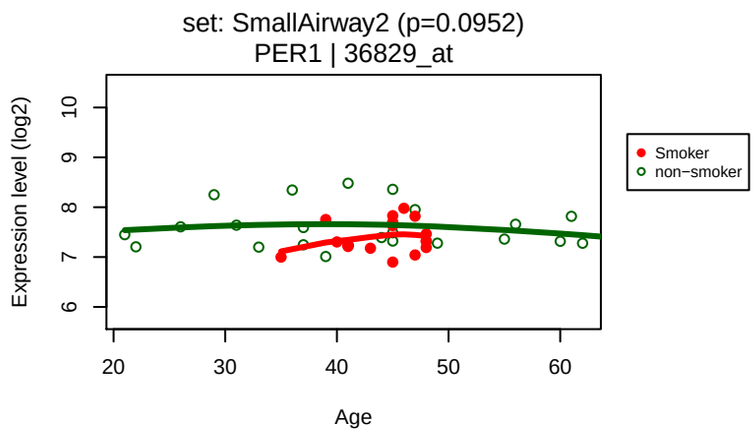
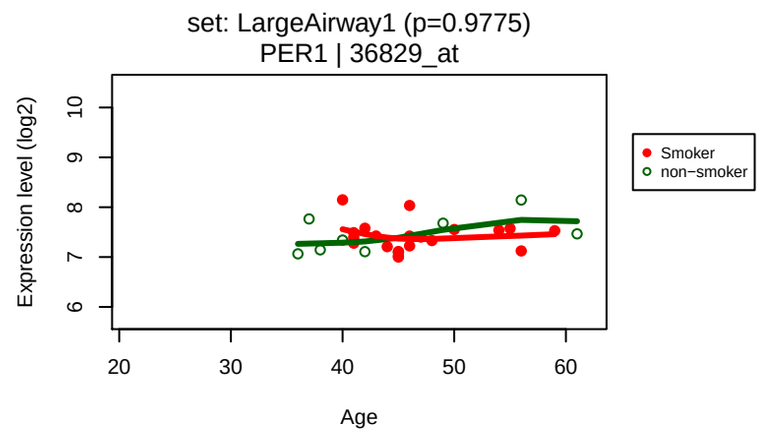

Gene expression of smokers and nonsmokers
gene: PER1 | 36829_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1033
p=0.0952
p=0.0619
p=0.9775
p=0.7517
p=0.016
p=0.7887
p=0.1833
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

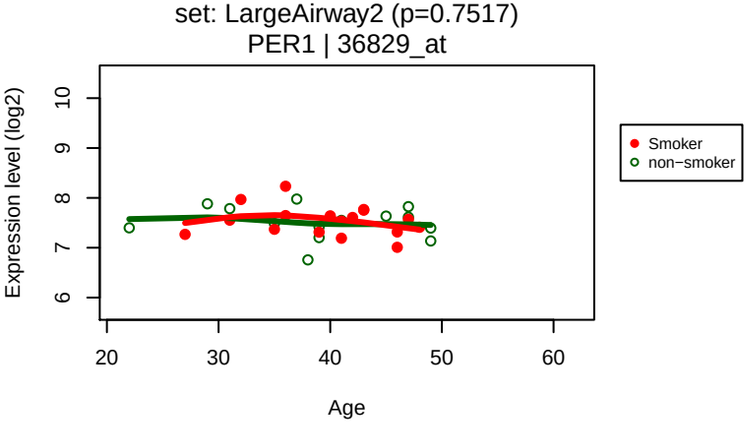

Gene expression ratio (smokers/nonsmokers)
gene: PER1 | 36829_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1033
p=0.0952
p=0.0619
p=0.9775
p=0.7517
p=0.016
p=0.7887
p=0.1833
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

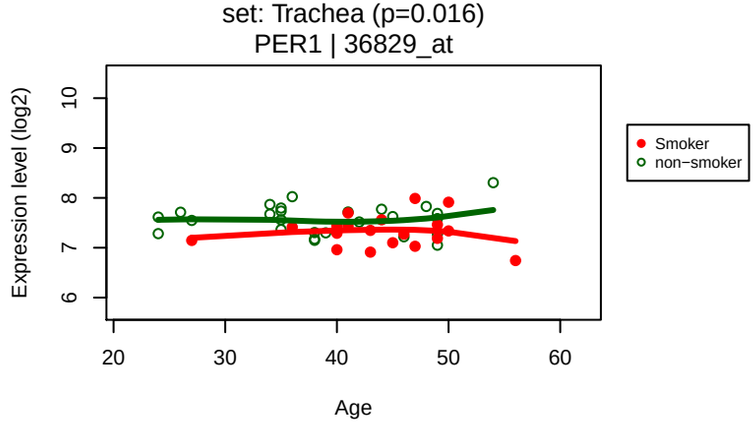

Gene expression ratio in Males
gene: PER1 | 36829_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.018
p=0.6979
p=0.2152
p=0.6183
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PER1 | 36829_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6093
p=0.9922
p=0.0014
p=0.1986
SmallAirway2
LargeAirway2
Trachea
Macrophage2

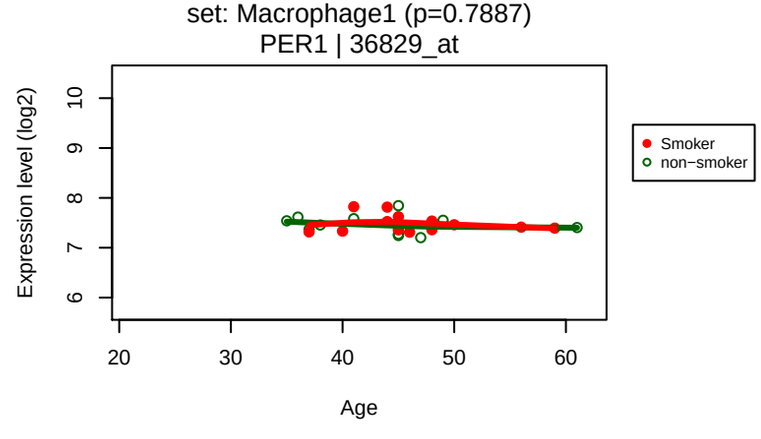

set: SmallAirway1 (p=0.1033)
PER1 | 36829_at
Expression level (log2)
Age
Smoker
non−smoker

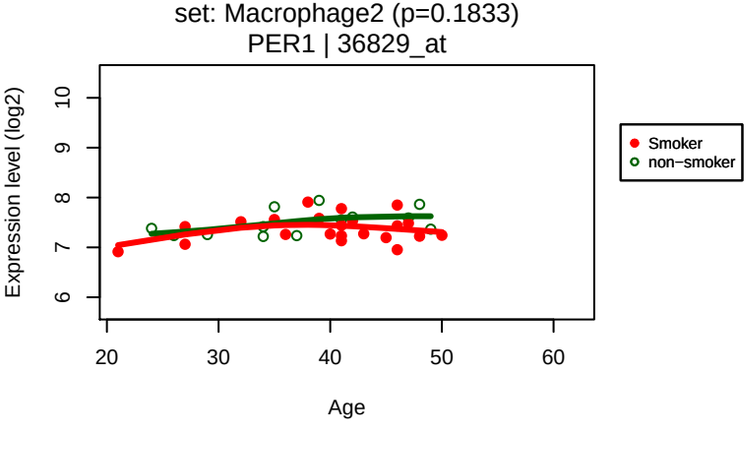

set: SmallAirway2 (p=0.0952)
PER1 | 36829_at
Expression level (log2)
Age
Smoker
non−smoker

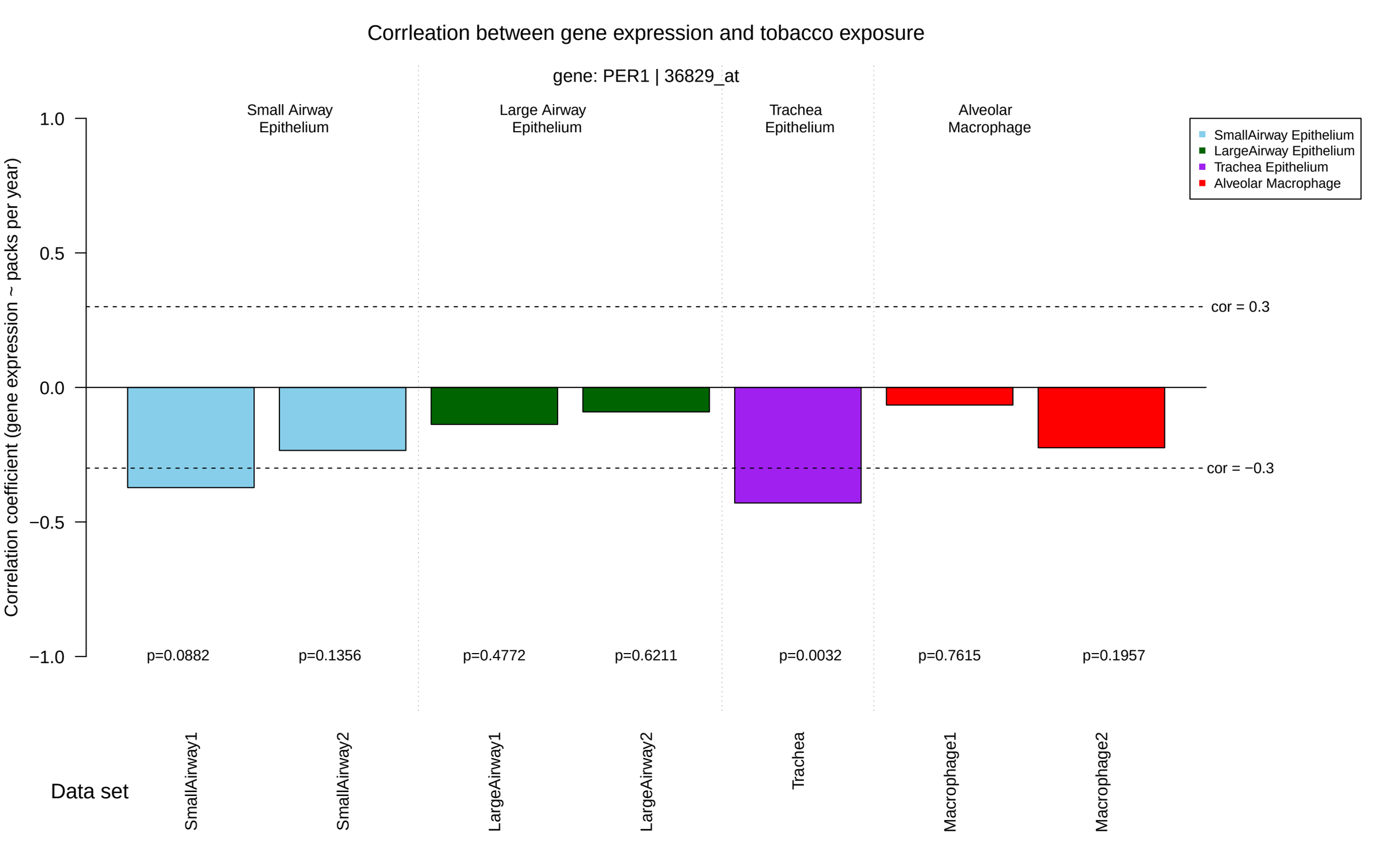

set: LargeAirway1 (p=0.9775)
PER1 | 36829_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7517)
PER1 | 36829_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.016)
PER1 | 36829_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7887)
PER1 | 36829_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1833)
PER1 | 36829_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PER1 | 36829_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0882
p=0.1356
p=0.4772
p=0.6211
p=0.0032
p=0.7615
p=0.1957
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2